Atlas-free cortical geometry: transport, spectral maps, and clinical inference

Three mathematical frameworks—point cloud conformal geometry, optimal transport, and functional maps—can now be combined into a fully parcellation-free pipeline for cortical morphometry, with mature Python implementations and feasible computation for clinical-scale datasets. The intersection of these methods opens a novel approach to brain asymmetry: representing each hemisphere as a point cloud, computing inter-hemispheric correspondence via spectral functional maps or Wasserstein distances, and testing associations with continuous clinical scores through distance-based regression. No existing paper applies all three frameworks jointly to cortical analysis, making this a genuine methodological contribution. The entire pipeline—from NIfTI binary masks to permutation-tested clinical associations—runs on standard hardware for **n=46 subjects at ~150k vertices per hemisphere** within hours.

Conformal flattening works directly on point clouds, not just meshes

The most important finding for point cloud cortical geometry comes from **Gary Choi and Lok Ming Lui** (CUHK/MIT), who developed conformal parameterization methods that operate directly on point clouds without requiring triangulated meshes. Choi, Ho, and Lui (2016, *SIAM J. Imaging Sci.*) achieved spherical conformal parameterization of genus-0 point clouds Lokminglui using a Moving Least Squares approximation of the Laplace-Beltrami operator, SIAM Harvard while Choi, Liu, and Lui (2022, *J. Sci. Comput.*) extended this to free-boundary disk parameterization using accumulated cotangent weights from local Delaunay triangulations over k-nearest-neighbor graphs. Springer arXiv The mathematical requirements are surprisingly minimal: sufficient point density, estimated normals, and ordered boundary points for open surfaces—no global mesh connectivity needed. arXiv arXiv

The **Beltrami coefficient** provides the key distortion measure. Meng and Lui (2018, *J. Sci. Comput.*) introduced the Point Cloud Beltrami Coefficient (PCBC), a discrete quasi-conformal distortion measure defined entirely on point clouds that converges to its continuous counterpart as density increases. CoLab +2 A Beltrami coefficient $|\mu| = 0$ indicates perfect conformality; values approaching 1 indicate degenerate mappings. PubMed Central For cortical surfaces, conformal flattening preserves angles but **not geodesic distances or areas**—the conformal factor λ can produce **5–10× area distortion** in deep sulci. Quasi-isometric alternatives exist (Choi, Giri, and Kumar 2022 developed adaptive area-preserving parameterization), Springer but no flattening can simultaneously preserve both angles and areas on the non-developable cortical surface.

The practical extraction pipeline is straightforward. From a NIfTI binary mask: load with nibabel, apply `skimage.measure.marching_cubes(mask, level=0.5)` to extract vertices and faces, apply the NIfTI affine to transform to world coordinates, then optionally discard faces for a pure point cloud. **PyMCubes** provides a `smooth()` function specifically for binary masks to eliminate staircase artifacts before extraction. GitHub Open3D then handles normal estimation, Poisson surface reconstruction, and ICP registration— Sigmoidal though it does **not** compute geodesic distances natively. Open3D Open3D For geodesics on point clouds, Mémoli and Sapiro's (2005, *SIAM J. Appl. Math.*) offset-band method Facundo-memoli or graph-based shortest paths through k-NN graphs are the established alternatives; the library `potpourri3d` provides efficient geodesic computation on meshes.

PointNet++ applied directly to cortical point clouds has already outperformed mesh-based methods. Vosylius et al. (2020, MICCAI ShapeMI) compared PointNet++, MeshCNN, and GraphCNN on 650 neonatal cortical surfaces from the Developing Human Connectome Project, finding PointNet++ with cortical features achieved the best prediction accuracy (MAE = 0.62 weeks for post-menstrual age). arXiv This demonstrates that point cloud representations are not merely adequate but competitive for cortical morphometry.

Optimal transport provides continuous asymmetry metrics, but exact methods are infeasible at cortical scale

Wasserstein distances have been applied to brain morphometry (Springer) since Su, Zeng, Wang, Lu, and Gu (2015, IPMI), who used Wasserstein-2 distance between conformal area-distortion measures on cortical surfaces to classify subjects by cognitive scores. Shi et al. (2016, CVPR) extended this with hyperbolic Wasserstein distances for longitudinal Alzheimer's tracking, achieving **76.7%** (TheCVF) **classification accuracy** distinguishing AD progression from normal aging. Gramfort, Peyré, and Cuturi (2015, IPMI) demonstrated optimal transport averaging directly on cortical surface triangulations with ~150k vertices, using geodesic distance on the folded cortex as the ground metric. (Springer) (ResearchGate) Gerber et al. (2023, *Med. Image Anal.*) introduced Unbalanced Transport Morphometry, which separates volume changes from location changes and increases statistical power for spatially dispersed pathology. (ScienceDirect) (PubMed Central)

No published work applies optimal transport specifically to inter-hemispheric asymmetry. This represents a clear gap. The natural formulation—computing Wasserstein distance between left and right hemisphere point clouds (with morphometric features like thickness and curvature as additional dimensions)—would yield a continuous, atlas-free asymmetry score per subject.

The critical computational constraint is that **exact OT and standard Sinkhorn are infeasible for 150k×150k problems**. The cost matrix alone requires ~180 GB in float64. Three scalable alternatives exist:

- **Sliced Wasserstein distance** (via POT: `ot.sliced_wasserstein_distance`): Projects point clouds onto random 1D directions, computes exact 1D Wasserstein, and averages. (arXiv) Runs in **1–5 seconds per pair** on CPU for 150k points with 200 projections. For 1,035 pairwise comparisons across 46 subjects, total compute is **~1–2 hours on CPU**. The sliced Wasserstein Gaussian kernel is provably positive definite (TheCVF) (Kolouri et al. 2016, CVPR), making it valid for kernel-based statistical methods. (TheCVF)
- **Sinkhorn divergences via GeomLoss** (Feydy et al., AISTATS 2019): Uses KeOps for CUDA kernel operations with **linear memory footprint**, handling millions of points. (GitHub +3) Demonstrated on brain tractograms at MICCAI 2019. (HAL) (Springer) Estimated **2–5 minutes per pair** on a V100 GPU for 150k points, totaling ~35–86 GPU-hours for all pairs. The `blur` parameter is anatomically meaningful (interpretable as spatial resolution in mm). (ADS) Now integrated into POT via `ot.solve_sample()`. (Pythonot)
- **FUGW (Fused Unbalanced Gromov-Wasserstein)**: Published at NeurIPS 2022 and available as `pip install fugw`, this library is **purpose-built for cortical surface alignment**. (arXiv) (GitHub) Its coarse-to-fine sparse solver scales explicitly to >150k vertices per hemisphere, computing a mapping in under 10 minutes on a V100 GPU. (github) It combines functional features with geometric structure, (ResearchGate) integrates directly with nilearn, and represents the most brain-specific OT implementation available. (Alexisthual)

The functional map C matrix encodes asymmetry more richly than any existing brain measure

The functional maps framework (Ovsjanikov et al. 2012, *ACM Trans. Graphics*) represents correspondence between two shapes as a **k×k matrix C** in the truncated Laplace-Beltrami eigenbasis, (ACM Digital Library) (SIGGRAPH) compressing a 150k×150k correspondence problem into typically a **100×100 matrix**. For two perfectly symmetric hemispheres, C would be diagonal (or near-diagonal with a characteristic "funnel" shape). The **off-diagonal Frobenius energy** $||C - \mathrm{diag}(C)||_F$ therefore provides a principled, continuous, atlas-free asymmetry metric that captures not just *that* hemispheres differ but *how* they differ across spatial frequencies.

This is strictly more informative than existing spectral approaches. **BrainPrint** (Wachinger et al. 2015, *NeuroImage*) compares only LBO eigenvalues (Shape-DNA) between structures, discarding all spatial correspondence information. ScienceDirect BrainPrint detected increased hippocampal asymmetry in Alzheimer's (Wachinger et al. 2016, *Brain*, n≈700 from ADNI), PubMed Central but eigenvalue comparison tells you shapes *differ* without indicating *where* or *how*. **BrainSpace** (Vos de Wael et al. 2020, *Commun. Biol.*) computes the same LBO eigenvectors as "gradients" but aligns them post-hoc via Procrustes rather than building explicit correspondence. PubMed Nature The functional map C matrix encodes the complete correspondence structure that both BrainPrint and BrainSpace discard.

Lombaert et al. (2015, IPMI) provide the most direct precedent, applying functional maps to cortical surfaces in their "BrainTransfer" framework. Springer They optimized a spectral transformation matrix (equivalent to C) between cortical surfaces, PubMed achieving matching accuracy comparable to FreeSurfer's spherical registration in **220 seconds vs. 3 hours**. More recently, **Mayer et al. (2024, *BMC Med. Imaging*)** used functional maps combined with graph convolutional networks for Alzheimer's disease grading directly from brain surface shape—the first application of functional maps to clinical brain shape classification.

The Python implementation path is clear. `robust_laplacian` (Sharp and Crane, SGP 2020 Best Paper) computes the LBO on both meshes and point clouds GitHub PyPI via `robust_laplacian.point_cloud_laplacian(points, n_neighbors=30)`, returning sparse matrices ready for eigendecomposition with `scipy.sparse.linalg.eigsh`. GitHub PyPI **pyFM** (Robin Magnet, INRIA) provides the complete functional maps pipeline: github descriptor computation (Wave Kernel Signature, Heat Kernel Signature), C matrix optimization, ZoomOut refinement (Melzi et al. 2019, SIGGRAPH Asia), and point-to-point map extraction. GitHub Computing k=100 eigenvectors for a 150k-vertex hemisphere takes **2–10 minutes**; the subsequent C matrix solve is effectively instantaneous.

A frequency-decomposed analysis is possible by examining different blocks of C. Low-frequency asymmetry (top-left block, corresponding to the first few eigenfunctions) captures large-scale hemispheric shape differences. High-frequency asymmetry (off-diagonal energy in higher-index blocks) captures fine-grained folding pattern differences. This multi-scale decomposition is unavailable from any scalar asymmetry measure.

Distance-based regression connects geometric representations to clinical outcomes

Four complementary statistical frameworks test associations between brain geometry distance matrices and continuous outcomes like HADS scores, all with valid permutation-based inference for n=46.

MDMR (Multivariate Distance Matrix Regression) is the most natural primary analysis. It directly regresses a distance matrix onto predictors by Gower-centering the squared distance matrix and decomposing its trace into variance explained by predictors versus residuals. The pseudo-F statistic tests whether HADS scores explain variation in brain geometry distances after controlling for covariates. Ponsoda et al. (2016) validated MDMR with cross-validation on **n=47 subsets**—virtually identical to the n=46 scenario—relating structural connectivity distances to cognitive abilities. MDMR handles covariates (age, sex, disease duration) through the design matrix X, with inference via **Freedman-Lane permutation** (the gold standard when covariates are present, validated by Winkler et al. 2014 in *NeuroImage*). The R `MDMR` package (McArtor et al. 2017) provides analytical p-values alongside permutation tests; a compact Python implementation requires only ~20 lines using scipy and numpy.

HSIC (Hilbert-Schmidt Independence Criterion) detects nonlinear dependence between kernel-embedded representations. Given a Gaussian kernel K_brain from brain geometry distances and K_hads from HADS score differences, HSIC = $(1/n^2)\mathrm{tr}(K_x H K_y H)$. Jejjohnson Sejdinovic et al. (2013) proved HSIC is formally equivalent to distance covariance, meaning `hyppo.independence.Dcorr` with precomputed distance matrices provides the same test. The Python **hyppo** library supports precomputed kernel/distance matrices with permutation inference and chi-squared fast approximation for n>20. Neurodata

RSA (Representational Similarity Analysis) correlates the brain geometry RDM with a clinical score difference matrix |HADS_i − HADS_j| via Spearman rank correlation over the 1,035 unique pairs. The **rsatoolbox** (Kriegeskorte lab) accepts custom RDMs and implements multiple comparison methods. Inference requires the Mantel permutation strategy (jointly permuting rows and columns) to respect the dependence structure of distance matrix entries.

Kernel Ridge Regression converts the distance matrix to a kernel via $K_{ij} = \exp(-d^2_{ij}/2\sigma^2)$ and predicts HADS scores via leave-one-out cross-validation, providing an effect size (R^2, MSE) interpretable as prediction accuracy. With n=46, the kernel matrix is 46×46, making computation trivial. The median heuristic for bandwidth selection is robust for small samples. Scikit-learn's `KernelRidge(kernel='precomputed')` accepts custom kernel matrices directly.

All methods require **10,000 permutations** for reliable p-values at publication quality (Winkler et al. 2016 showed 5,000 is minimum for 95% CI [0.044, 0.056] around p=0.05). With 46×46 matrices, each permutation test completes in seconds.

A complete pipeline runs on standard hardware in under two days

The entire workflow from NIfTI masks to clinical inference is computationally tractable with the following estimated budget:

Phase 1 — Preprocessing (~8–23 hours on 8 CPU cores for 92 hemispheres): Load surfaces, compute LBO eigenbasis (k=100) via robust_laplacian + scipy.eigsh, extract morphometric descriptors. This is embarrassingly parallel across subjects.

Phase 2 — Pairwise distances (choose one or combine):

Method	Time (1,035 pairs)	Memory	Output
Functional map C matrices	~1–2 hours CPU	100×100 per pair	Dense correspondence + asymmetry metric
Sliced Wasserstein (POT)	~1–2 hours CPU	~5 MB per pair	Scalar distances
Sinkhorn divergence (GeomLoss)	~35–86 GPU-hours	~2 GB GPU	Approximate OT distances
FUGW sparse (full alignment)	~170 GPU-hours	~4–8 GB GPU	Dense transport plans

Phase 3 — Statistical analysis (minutes): Construct 46×46 distance/kernel matrices, run MDMR + HSIC + Mantel + KRR with 10,000 permutations each.

The critical software stack: **nibabel** (NIfTI I/O), **scikit-image** (marching cubes), **Open3D** (point cloud processing, normal estimation, ICP), Open3D Open3D **robust_laplacian** (LBO on meshes and point clouds), GitHub **scipy.sparse.linalg.eigsh** (eigendecomposition), SciPy **pyFM** (functional maps), github **POT** (sliced Wasserstein), ACM Digital Library JMLR **GeomLoss** (scalable Sinkhorn), Kernel-operations **FUGW** (brain-specific OT alignment), OpenReview GitHub **hyppo** (HSIC/distance correlation), **scikit-bio** (Mantel test), and **rsatoolbox** (RSA). All are pip-installable.

The key bottleneck is eigendecomposition at 150k vertices (2–10 minutes per hemisphere), mitigated by shift-invert mode (`sigma=0, which='LM'`) in scipy. Full OT cost matrices at 150k×150k (~180 GB) are categorically infeasible; sliced Wasserstein and GeomLoss's linear-memory Sinkhorn are the viable alternatives. Pythonot

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Epilepsy cortical morphology beyond parcellation: what exists and what's missing

Current epilepsy morphometry relies heavily on FreeSurfer parcellation-based approaches: the Morphometric Analysis Program (MAP) for detecting focal cortical dysplasia, vertex-wise cortical thickness analyses on fsaverage templates, and voxel-based morphometry. These methods require atlas registration, impose parcellation boundaries, and lose continuous geometric information.

No published work applies optimal transport, functional maps, or point cloud geometry specifically to epilepsy cortical morphology. The closest relevant work is Gerber et al.'s (2023, *Med. Image Anal.*) Optimal Transport Features for morphometric population analysis in Alzheimer's, which the authors note is applicable to any morphometry pipeline. PubMed Central PubMed Their key finding—that OT-based features require **smaller sample sizes** to detect differences than standard VBM/TBM—is directly relevant to clinical epilepsy studies where n is typically small. ScienceDirect UNC CDR

The proposed framework fills three specific gaps in epilepsy neuroimaging. First, continuous asymmetry quantification: instead of comparing parcellated cortical thickness between hemispheres (which is standard in focal cortical dysplasia detection), Wasserstein distances or functional map off-diagonal energy provide a single continuous measure capturing both the magnitude and spatial pattern of asymmetry without parcellation artifacts. Second, atlas independence: point cloud methods bypass the registration-to-template step that can introduce systematic bias in patients with significant structural abnormalities. Third, whole-hemisphere characterization: distance-based regression (MDMR) with clinical scores like HADS captures global morphological relationships that parcellation-based methods miss by testing each region independently.

Conclusion

The three frameworks converge naturally through the LBO eigenbasis: conformal geometry uses it for flattening, functional maps use it for correspondence, and even optimal transport benefits from spectral embeddings for computational tractability (as in FUGW's Landmark MDS approximation). The functional map C matrix emerges as the most theoretically principled asymmetry measure—more informative than BrainPrint's eigenvalue comparison and more geometrically grounded than BrainSpace's gradient alignment. Sliced Wasserstein distance provides the most computationally efficient alternative (arXiv) when dense correspondence is unnecessary. The statistical layer (MDMR + HSIC with Freedman-Lane permutation) is well-validated for exactly this sample size regime. Applying this combined framework to epilepsy cortical morphology—particularly continuous asymmetry quantification correlated with anxiety/depression via HADS—would constitute a genuinely novel methodological contribution to clinical neuroimaging.